Embedded Controller

System Tick Timer

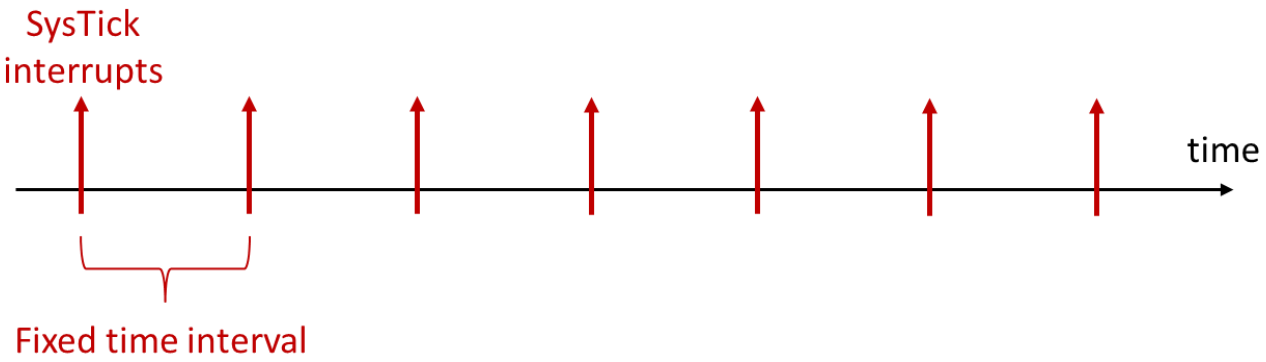
Young-Keun Kim
School of Mechanical & Control Engineering

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Last modified: 2025.8

System Tick Timer (SysTick)

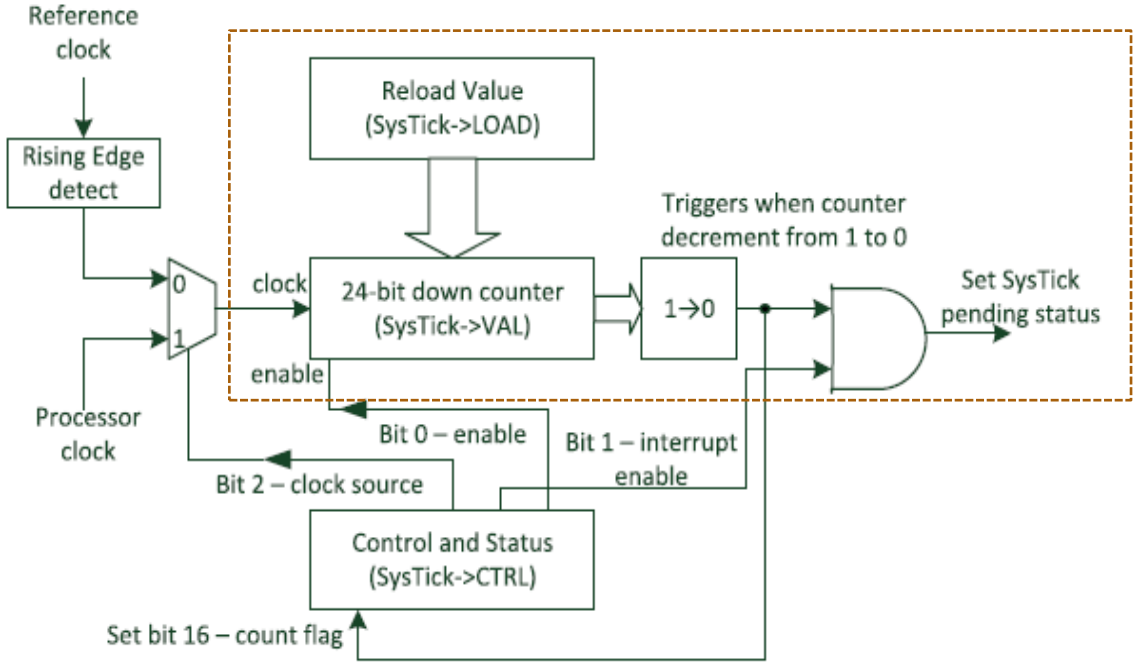
- **System Tick (SYSTICK)**
  - Cortex M3/M4 have a built-in system tick timer called SysTick (24-bit timer)
  - Generate regular timer interrupts
    - Periodic interrupt
    - Time measurement
    - OS timekeeping

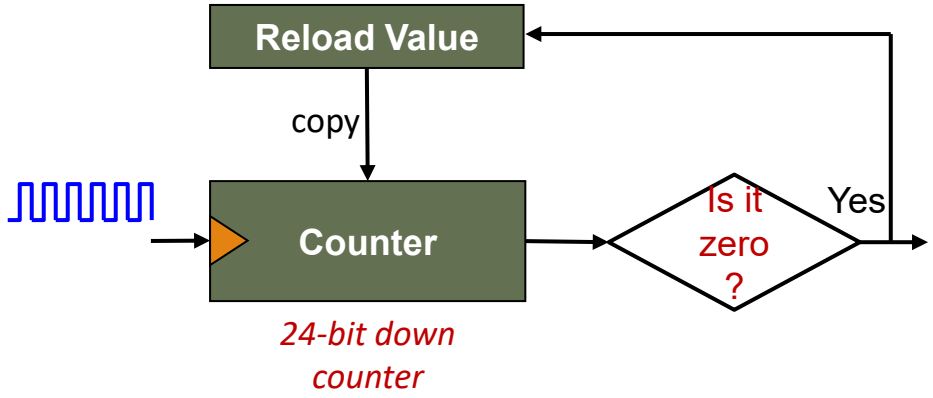



```
void SysTick_Handler(void){
  ...
}
```

System Tick Timer (SysTick)

- **SysTick Timer**
  - 24-bit down counter

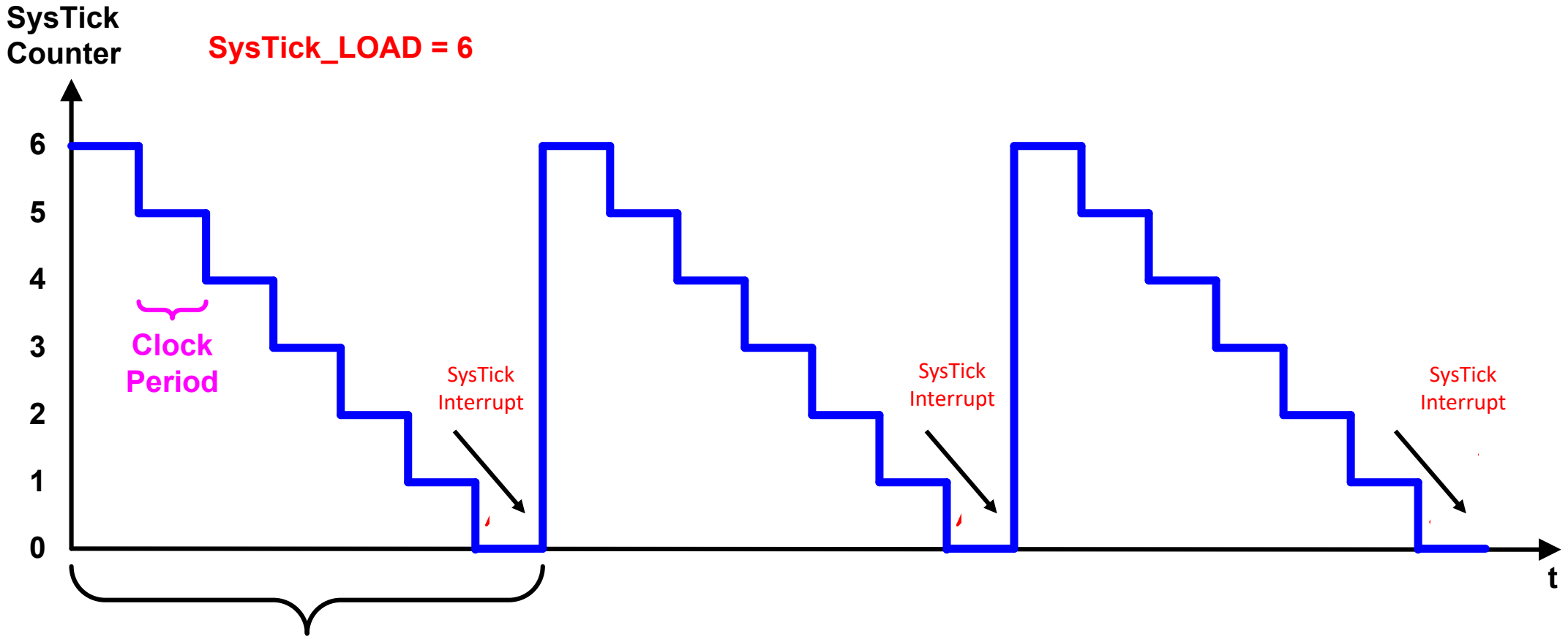




Refer to Program manual [PM]

System Tick Timer (SysTick)

Example: down counting from 6 to 0



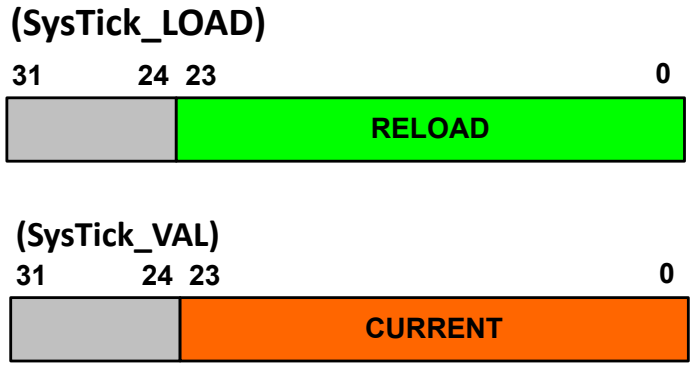
SysTick Interrupt Time Period = (SysTick_LOAD + 1) × Clock Period = 7 × Clock Period

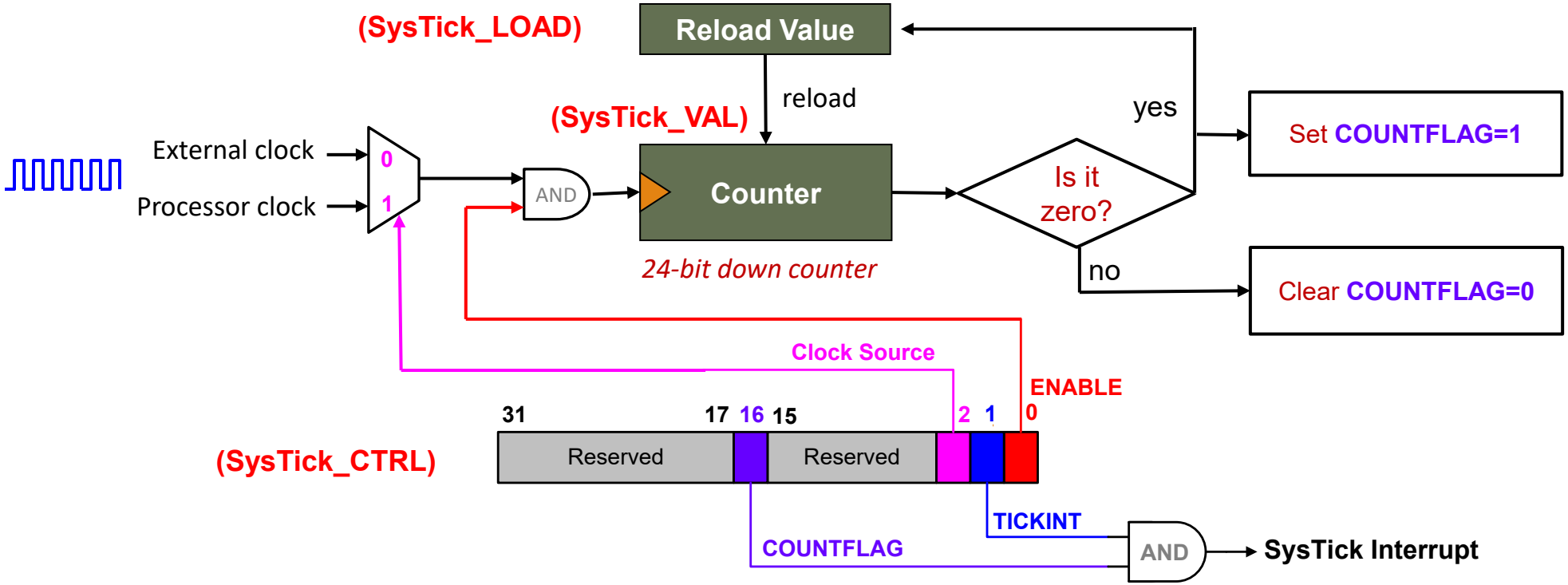
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System Tick Timer (SysTick)

Registers of SysTick

- **SysTick_CTRL**: SysTick control and status register
- **SysTick_LOAD:** SysTick reload value register (24 bits, ~0x00FF FFFF)
- **SysTick_VAL**: SysTick current Value (always initially reset before use)





System Tick Timer (SysTick)

Calculating Reload Value

- Example
  - Suppose clock source = 84MHz
  - Goal: SysTick Interval = 10ms
  - How to choose RELOAD value?

$$Tick\ Period = (Reload + 1) * Clock\ Period$$

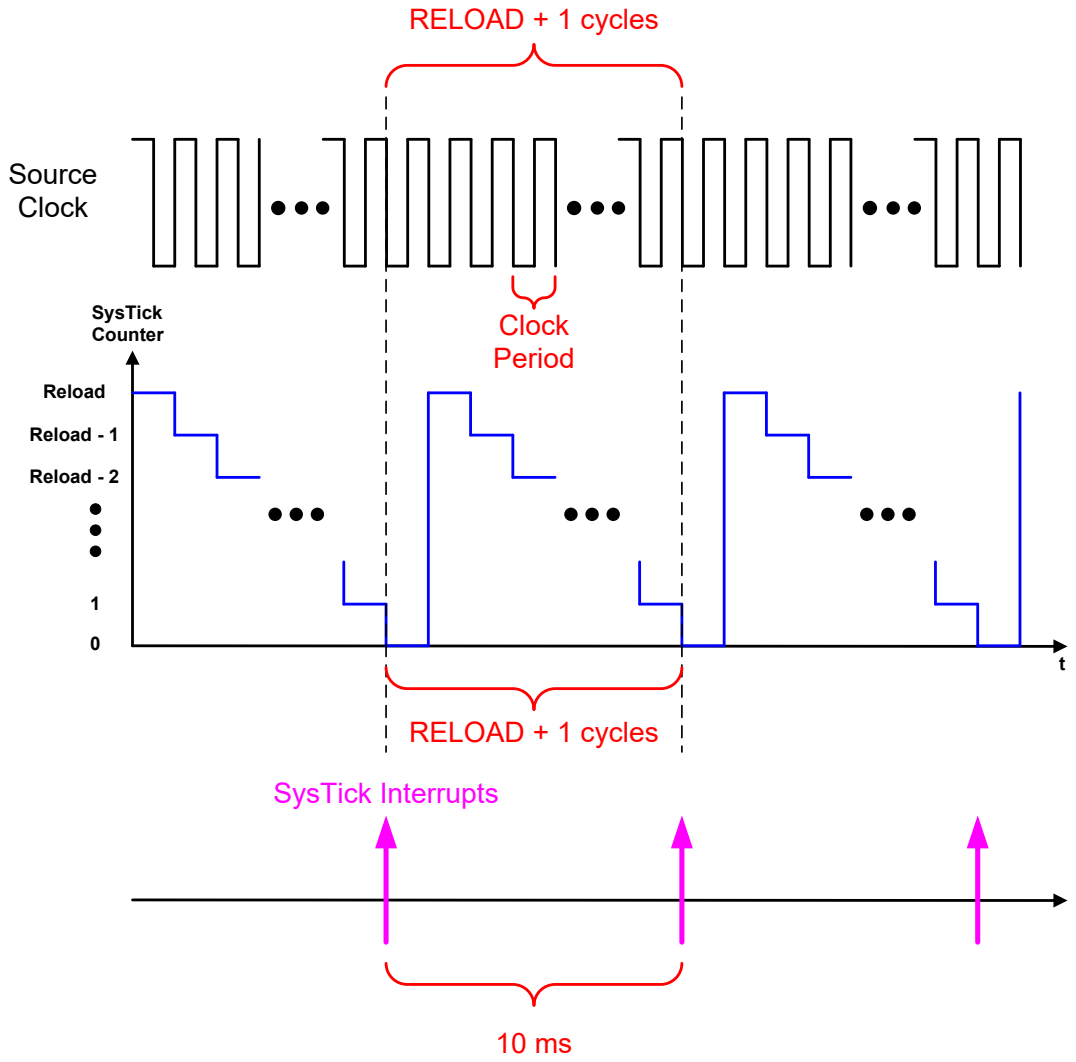
$$Reload = \frac{Tick\ Period}{Clock\ Period} - 1$$

$$= \boldsymbol{Tick\ Period} * Clock\ Freq. - 1$$

$$= 0.01s \times 84MHz - 1$$

$$= 840{,}000 - 1$$

$$= 839{,}999$$



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System Tick (SysTick) Interrupt Register Description

Embedded Controller

Prof. Young-Keun Kim

SysTick Register

- **SysTick (STK or SysTick)**
  - **Clock Control and Status**
    SysTick_CTRL
  - **SysTick Reload Value**
    SysTick_LOAD
  - **SysTick Current Value**
    SysTick_VAL
  - **SysTick Calibration**
    SysTick_CALIB

For SysTick, refer to Programming Manual
NOT Reference manual

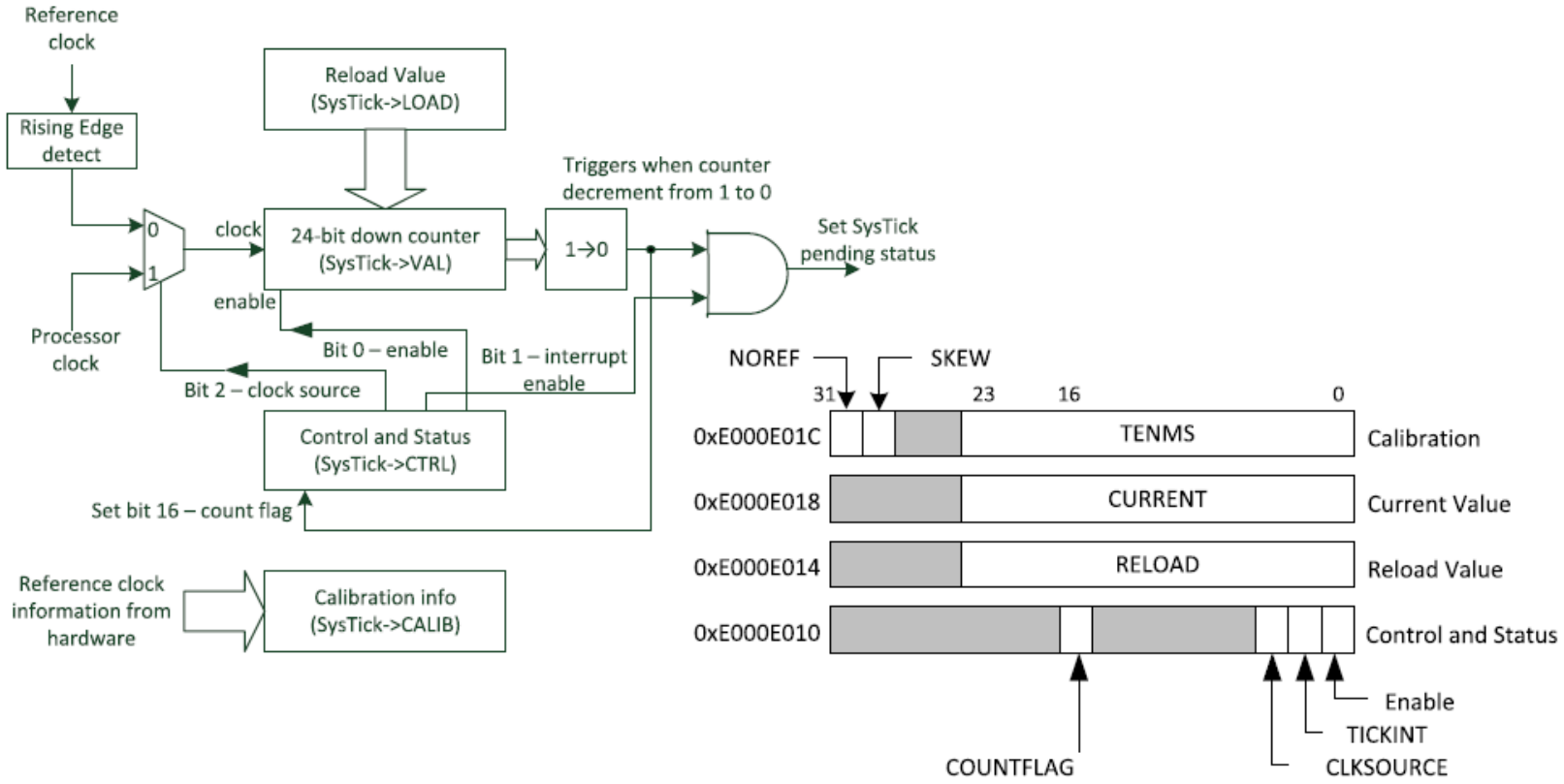


FIGURE 9.15

A simplified block diagram of SysTick timer

<YIU, Ch.9, pg. 314>

SysTick Register

(RCC system clock)

Assume: PLL, HCLK= 84MHz

(System Tick Configuration)

1. Disable SysTick TImer	**SysTick->CTRL: ENABLE**=0
2. Choose clock signal: Processor clock	**SysTick->CTRL: CLKSOURCE** = 1
3. Choose to use Tick Interrupt (timer goes 1->0)	**SysTick->CTRL: TICKINT** = 1
4. Set Reload Counting value (24-bit)	**SysTick->LOAD: RELOAD** = ReloadValue
5. Enable SysTick Timer	**SysTick->CTRL: ENABLE**=1
6. (option) Read or Clear current counting value	**int val= (SysTick->VAL)**

- Write clears value in (SysTick->VAL)

(NVIC Configuration)

- NVIC SysTick Interrupt priority and Enable

SysTick (System Tick Timer) Register

- **System Tick Configuration**

1. Disable/Enable SysTick Timer: **SysTick->CTRL: ENABLE**
2. Choose clock signal: System clock or ref. clock(STCLK)**:** **SysTick->CTRL : CLKSOURCE**
3. Choose to use Tick Interrupt (timer goes 1->0): **SysTick->CTRL: TICKINT**

4.5.1 SysTick control and status register (STK_CTRL)

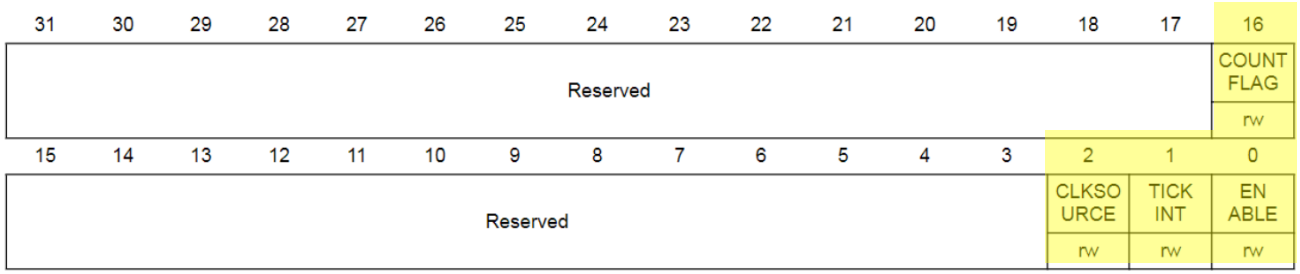


Table 9.7 SYSTICK Control and Status Register (0xE000E010)

Bits	Name	Type	Reset Value	Description
16	COUNTFLAG	R	0	Read as 1 if counter reaches 0 since last time this register is read; clear to 0 automatically when read or when current counter value is cleared
2	CLKSOURCE	R/W	0	0 = External reference clock (STCLK) 1 = Use core clock
1	TICKINT	R/W	0	1 = Enable SYSTICK interrupt generation when SYSTICK timer reaches 0 0 = Do not generate interrupt
0	ENABLE	R/W	0	SYSTICK timer enable

SysTick (System Tick Timer) Register

- **System Tick Configuration**

  4. Configure Reload Counting value (24-bit):

SysTick->LOAD: RELOAD = Reload

$$Reload = \text{Interval[s]} * CLK\ \text{Freq[Hz]} - 1$$

4.5.2 SysTick reload value register (STK_LOAD)

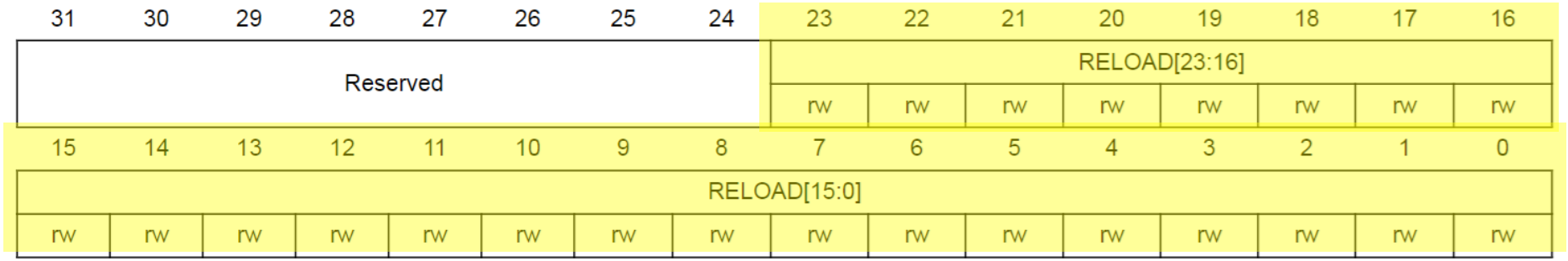


Table 9.8 SYSTICK Reload Value Register (0xE000E014)

Bits	Name	Type	Reset Value	Description
23:0	RELOAD	R/W	0	Reload value when timer reaches 0

SysTick (System Tick Timer) Register

- **System Tick Configuration**

  6. Read or Clear current counting value: **int val= (SysTick->VAL)**

4.5.3 SysTick current value register (STK_VAL)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved								CURRENT[23:16]							
								rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CURRENT[15:0]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 23:0 **CURRENT:** Current counter value

The VAL register contains the current value of the SysTick counter.

Reads return the current value of the SysTick counter.

A write of any value clears the field to 0, and also clears the COUNTFLAG bit in the STK_CTRL register to 0.

Example Code: SysTick Configuration

- **Example code: Using SysTick Initiation**

```
void SysTick_init(uint32_t tickPeriod) {
    SysTick->CTRL = 0;            // Disable SysTick
    SysTick->LOAD = tickPeriod*SYSCLK - 1;    // Set reload register,  SYSCLK=84,000,000 [Hz]
    // Set interrupt priority of SysTick to least urgency (i.e., largest priority value)
    NVIC_SetPriority(SysTick_IRQn, 15);
    SysTick->VAL = 0;             // Reset the SysTick counter value
    // Select processor clock: 1 = processor clock; 0 = external clock
    SysTick->CTRL |= SysTick_CTRL_CLKSOURCE;
    // Enables SysTick interrupt, 1 = Enable, 0 = Disable
    SysTick->CTRL |= SysTick_CTRL_TICKINT;
    // Enable SysTick
    SysTick->CTRL |= SysTick_CTRL_ENABLE;
    // Interrupt Priority
    NVIC_SetPriority(SysTick_IRQn, 16); // Set Priority to 16
    NVIC_EnableIRQ(SysTick_IRQn);      // Enable interrupt in NVIC
}
```